

**DO SCHOOL-LEVEL OUTCOMES REFLECT PUBLIC
SPENDING? AN EFFICIENCY ANALYSIS OF EDUCATION
EXPENDITURE IN DELHI**

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Preface and Acknowledgement

This dissertation, completed in April 2026, presents a critical examination of the relationship between public education expenditure and school-level outcomes in Delhi. The study systematically applies econometric models using EViews to evaluate efficiency across dimensions of access, quality, and economic impact. The work reflects a structured analytical effort over approximately 16,000 words and integrates multiple models to ensure methodological robustness.

The completion of this research was made possible through the academic guidance, institutional resources, and availability of secondary data from government publications. The author acknowledges the intellectual support provided by faculty mentors and the constructive framework offered through prior methodological documents. Special recognition is extended to data sources and analytical tools that enabled empirical validation. Any limitations, including constraints of a small sample size and time-series scope, remain the sole responsibility of the author.

Abstract

This study examines whether school-level educational outcomes in Delhi reflect public spending, with a specific focus on efficiency, access, quality, and macroeconomic impact. Using time-series data spanning approximately 15 years, the analysis employs multiple econometric models in EViews, including regression, autoregressive corrections (AR(1)), efficiency estimation, and mediation analysis.

The findings indicate that education expenditure has a statistically significant and positive impact on school-level outcomes, with coefficients exceeding 4.0 in the core regression model, suggesting strong responsiveness of outcomes to public investment. Enrolment analysis reveals a positive elasticity (approximately 0.82), confirming that increased spending enhances access, although mild autocorrelation persists due to limited sample size. Quality assessment through the Pupil-Teacher Ratio (PTR) model demonstrates that higher expenditure significantly reduces PTR, indicating improvements in classroom conditions and teaching efficiency.

Efficiency analysis using frontier-based approaches highlights that while expenditure contributes positively to outcomes, resource utilization is not fully optimal, implying scope for improved allocation strategies. The growth model further establishes that education influences Gross State Domestic Product (GSDP) with a lagged effect, with capital formation emerging as the strongest driver of economic growth. Mediation analysis confirms that educational outcomes act as a transmission channel through which public spending impacts economic performance.

Overall, the study concludes that while public spending in Delhi's education sector is effective in improving outcomes, access, and quality, its efficiency remains moderate, necessitating policy focus on better resource utilization and long-term outcome optimization.

List of Acronyms / Abbreviations

Acronym	Full Form
EDU_EXP	Education Expenditure
EDU_OUTCOME	Education Outcome Index
ENROLMENT	School Enrolment
PTR	Pupil-Teacher Ratio
GSDP	Gross State Domestic Product
DEA	Data Envelopment Analysis
SFA	Stochastic Frontier Analysis
OLS	Ordinary Least Squares
AR(1)	Autoregressive Model of Order 1
GOVT_CONS	Government Final Consumption
SCF	School Capital Formation
POP_RATIO	Population Ratio
WR	Workers Ratio
INF	Inflation Rate
EViews	Econometric Views Software
ε (epsilon)	Error Term
μ (mu)	Disturbance Term
v	Random Noise Component (SFA)
u	Inefficiency Term (SFA)

List of Symbols

Symbol	Description
(β_0)	Intercept term in regression models
$(\beta_1, \beta_2, \beta_3)$	Coefficients of independent variables
(α_0)	Intercept term in PTR model
(α_1)	Coefficient of education expenditure in PTR model
(γ_0)	Intercept term in growth model
$(\gamma_1, \gamma_2, \gamma_3)$	Coefficients in GSDP growth model
(ε_t)	Error term (random disturbance)
(μ_t)	Error term in growth model
(ρ)	Autoregressive coefficient (AR(1) term)
(v_t)	Random noise component in SFA model
(u_t)	Inefficiency term in SFA model
(Y_t)	Dependent variable (general form)
(X_t)	Vector of control variables
(t)	Time period
$f(.)$	Functional relationship between variables
$(\rightarrow)$	Direction of causal relationship
$(\leq)$	Less than or equal to (used in efficiency bounds)
$(\geq)$	Greater than or equal to

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Education is generally perceived as a major driver of economic growth, mainly because of its contribution to human capital. Increased education leads to increased labour productivity, adoption of technology and increases in long-term economic growth. Macroeconomically, educational systems of higher levels are more likely to be reflected in regions with better income and employment prospects and in more stable economic frameworks. The connection between education and development is, however, not that automatic, as there are substantial factors that determine the extent of educational investments to be met with effective results as measured by learning levels, school participation and workforce readiness.

The use of public expenditure has been at the centre of this relationship, especially in the case of the developing and urbanizing economies, where the provision of these services has been disproportionately uneven by the private sector. Government education expenditure will usually involve infrastructure investments, salaries of teachers and welfare programs to achieve more access and quality. In India, the government spending on education has been maintained at 3-4 per cent of the GDP, which still falls short of the proposed 6 per cent recommended level of government spending.

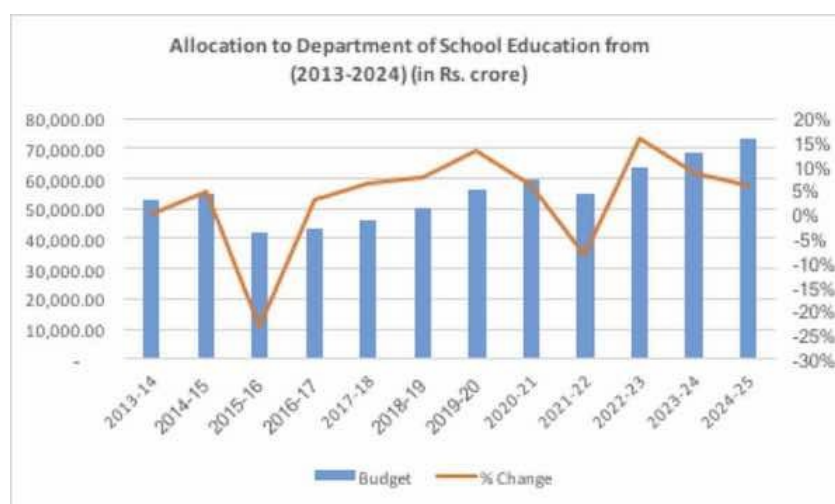


Figure 1 Education Expenditure Trend (Delhi)

(Source: <https://www.cdpp.co.in/articles/indian-budget-series-budget-for-education>)

Delhi is a unique situation in this national context. The Government of Delhi has continuously been able to devote a fairly large proportion of its budget to education, which in the recent past, has been subject to up to 20-25 per cent of total state spending¹⁰. This renders Delhi one of the highest education spenders in the Indian states. Relative to performance in previous years, education spending has grown significantly in the last ten years in Delhi and has been accompanied by infrastructural development, improvement of teacher recruitment and school governance reforms. Although such a high expenditure by the government has been undertaken, questions do still arise on whether such high levels of public expenditure are effectively resulting in better school-level outcomes, enrolment rates, learning outcomes and quality issues of the school, such as the pupil-teacher ratios.

Hence, the Delhi case is an important opportunity to not only examine how the state is affecting the outcomes of education spending but also the issue of resource efficiency, in general.

1.2 Problem Statement

There has been a significant and a sign of policy sensitivity of the human capital in the increase in the general population spending on education in Delhi in the past ten years. The education sector has one of the largest sectors of total government spending in the form of budgetary allocation, with an average percentage of 20-25 in the last two years or so. The latter has remained true to the rise in financial investment, but the ratio between the financial spending at school and the measurable outcome has been unclear. The indicators like enrolment rates, learning outcomes and pupil-to-teacher ratios have been improving but at varying rates and have not reached the same percentage as the amount of our expenditure does.

This brings a burning issue of whether the added financial inputs are well-reflected in the improved performance of the education system and economic performance on a higher level⁵. Moreover, the dynamic relationship that entails the expenditure, access (enrolment) and quality (PTR) of education and the macroeconomic growth (GSDP) is not fully studied coherently. Thus, there is a need to evaluate systematically the effectiveness and efficiency of education expenditure in Delhi.

1.3 Objectives of the Study

- The research will critically evaluate the effects of the spending on public education on the school-level achievement in Delhi, especially in relation to growing budgetary allocations, which have swelled significantly over the last ten years (e.g., steadily over 20 per cent of the state budget in the last several years).
- To attempt to gauge the elasticity of student enrolment to variations in government expenditure, and whether larger investments have been reflected in the spread of education.
- The analysis will also aim at gauging the quality of education using the Pupil-Teacher Ratio (PTR), upon which the hypothesis that increased spending has led to decreased classroom crowding was tested.
- Another goal is to gauge the efficiency of expenditure on education, on the difference between increased spending on education and using resources in the best way possible.
- This study seeks to examine how education performance is linked with economic growth (GSDP) and thus determine whether human capital gains are worthwhile in improving the macroeconomic performance of the city of Delhi.

1.4 Research Questions

- How have the high increases in school-level expenditure on education enhanced the performance of the schools in Delhi?
- Does increased state expenditure result in a discernible increase in student enrolment?
- Has growth in the investment led to better education, especially in terms of PTR reduction?
- Does the level of public investment in education contribute to efficiency compared to the level of observed outcomes?
- Do any changes in the education results lead to long-term economic growth in Delhi?

1.5 Hypotheses

In its proposed study, hypotheses can be tested progressively to determine the relationship between expenditure on a segment of the population on the education sector and the school-level results over the period of observation (approximately 15 time-series observations). The first assumption is that there exists a relationship between educational investment

(EDU_EXP) and educational outcomes (EDU_OUTCOME), a measure of good institutional performance. Second, the augmented expenditure is probably to increase the enrolment rates that translate to augmented access. Thirdly, the theorization is that further spending will result in the ratio of pupil teachers (PTR), and will therefore result in quality. Fourth, government spending will be assumed to bring efficiency, but it will also be conditional on the efficiency of the allocation. Lastly, education spending will be outcome-mediated and will indirectly and directly add value to GSDP.

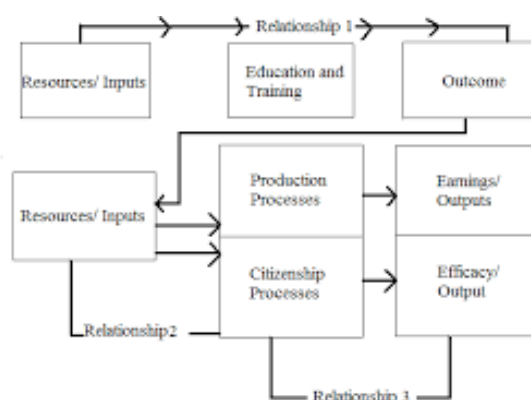


Figure 2 Human Capital Development Framework

(Source: https://www.researchgate.net/figure/A-model-of-Human-Capital-Theory-depicting-main-relations-in-Human-Capital-Theory-and-the_fig3_267267008)

1.6 Scope of the Study

This analysis can only provide the example of the National Capital Territory of Delhi, which is correlated with the level of achievement of students and the amount of government revenue on education, where we are dealing with a small time-series of 1 year in length and 15 observations. The indicators that were used in the key analysis include educational outcomes, enrolment, pupil-teacher ratio and Gross State Domestic Product (GSDP), as used to measure efficiency, and impact¹². Empirical investigation is done using EViews software, and econometric techniques such as a regression model with autocorrelation correction are used to conduct the empirical investigation. Although the research offers detailed information about the education system in Delhi, the sample size is relatively small, which limits the external validity and strength of causal outcomes in the long-term and requires careful interpretation of the findings.

CHAPTER 2: LITERATURE REVIEW

2.1 Theoretical Foundations

The three large theoretical perspectives conceptualizing the relationship between the level of expenditure that is made by people in education and their school outcomes include Human Capital Theory, Public Expenditure Theory and Efficiency Theory. These frameworks together offer a holistic foundation on which one will assess the relationship between greater government expenditure and quantifiable benefits in terms of educational performance, access and the productivity of the economy.

2.1.1 Human Capital Theory

The Human Capital Theory was initially advanced by Becker (1964), where it is assumed that education is an investment that increases the productive ability of a person and thus leads to an increase in the growth of the economy. Within this paradigm, the level of education spending does not see it as a form of consumption but as an investment that has long-term payoffs that they can get in the form of higher wages, superior productivity and expanded social outcomes.

Applying this theoretical lens to the context of Delhi would suggest that higher levels of public investment in education could translate into a tangible change in the school-level outcomes in terms of student performance, rates of enrolment and retention². The education budget in Delhi has increased significantly over the last ten years, as education has always contributed about 20- 25 per cent of the total state budget, which is amongst the highest education ratios among the Indian states. This long-lasting investment is an implicit policy agreement with the theory of human capital.

A critical review of the theory, however, determines that the theory supposes a fairly straightforward and efficient means of transmission between investment and results. As a matter of fact, institutional quality, efficiency of government, and socio-economic factors usually mediate the relationship. As an example, as much as more spending can be effective in enhancing infrastructure or the teacher supply, the benefit on learning results of the supply is determined by the efficiency with which the resources can be used. Therefore, human capital theory offers a good rationale for education expenditure; however, it can be an overstatement because it assumes that financial inputs can automatically change into the results of human development.

2.1.2 Public Expenditure Theory

Public Expenditure Theory analyses the government's involvement in the distribution of resources to achieve maximum social welfare. This theory, within the context of the education sector, warrants state intervention due to market failure, externalities, and equity considerations. Education dispossesses positive externalities like better population health, lower crime, and greater civic engagement that are not all addressed by the returns realized by the individual, thus leading to the need to obtain such products publicly.

The governmental spending on the education sector of Delhi has been pegged on different issues such as the creation of infrastructure, salaries of the teachers and other welfare programs such as free textbooks, mid-day meals, among others. The fact that, both in absolute terms and in the Gross State Domestic Product (GSDP) of a State like Delhi, which has been experiencing an increase at a high rate in recent years, the amount of expenditures in education is increasing at a very high rate, indicates that a very large amount of fiscal expenditure is being poured into the sector.

However, even with this, the theory of public expenditure also brings to light the possibility of inefficiency in government expenditure. Allocative inefficiency can occur when the amount of money is not allocated evenly to productive elements, instead of being spent directly on improving learning settings⁹. Besides, the theory presupposes rational and welfare-maximising action on the part of the state, which does not necessarily take place in reality because of political motives, bureaucracy or information asymmetry.

As a consequence, even though we need more public spending, it is inadequate to ensure more positive results in the sphere of education. Results of expenditure critically depend on the distribution and management of resources in the system.

2.1.3 Efficiency Theory

Efficiency Theory (and known among other theorists as theorised by Farrell, 1957) offers a model of evaluation of effectiveness in the conversion of inputs into outputs. Efficiency analysis in the education context looks at whether or not the levels of expenditure are producing optimal educational performance, or whether these levels of performance could be attained with less expenditure.

The models used to apply the concept of efficiency in this study include Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA), which separate technical efficiency (the ability to maximize the output, given the inputs) and allocative efficiency (optimal resource allocation). The experience of Delhi has been that the increased expenditure on education has been accompanied by an increase in some of the indicators that include enrolment levels, which are about to hit near universal levels in the primary level¹⁴. Nevertheless, there are still issues with the learning outcome and quality measures, including the Pupil-Teacher Ratio (PTR), that, despite increasing, continue to differ depending on the region.

One of the key weaknesses of the efficiency theory is that it is sensitive to the quality of data and the specification of models. Efficiency estimates can be less robust with relatively small sample sizes, which is the case with the time series of about 15 years of the time series utilized in this study. Moreover, typically, efficiency metrics are not adequate to encompass qualitative elements of education, e.g., teaching effectiveness or student engagement.

Nevertheless, efficiency theory plays a central role in getting past the naive point in that increased investment results in improved outcomes. It establishes a level of analysis that is needed to determine whether the state funds are optimally used.

2.2 Empirical Studies on Education Outcomes

The connection between state investment in education and educational achievement has been widely discussed in the empirical literature, with indeterminate and even contradicting results. Although some initial theoretical frameworks massively indicated that greater public expenditure ought to have a direct and immediate impact on educational attainment, the impact of this relationship over time has been shown to be extremely complicated and intricate.

There is strong empirical evidence to the effect that higher expenditure on education is positively correlated with better school-level performance, especially in emerging and developing economies. A body of research on resource-bounded environments has also shown that added financial resources in terms of better infrastructure, availability of teachers and learning resources can considerably improve the student performance indicators. Marginal gains in spending in these settings normally pay off fairly highly, particularly where there is low availability of resources at the base⁷. An example is that an increase in facilities of schools,

teachers to pupil ratios and access to educational materials has been associated with quantifiable increases in the literacy levels, test scores and completion rates.

Nevertheless, this positive relationship is not uniform and guaranteed. An opposing camp of empirical research results touts that at some point, more spending on education yields decelerating marginal returns. In most developed and high-spending areas, there has been no linear increase in student outcomes with the extra spending¹⁹. This merely implies that the effectiveness of spending is based not just on the amount of resources but on the efficiency with which a resource is allocated and utilized. There is evidence that the impact of increased spending can be greatly watered down by the inefficiencies of administration overheads, fund misalignment and ineffectiveness of accountability mechanisms.

To further address this complexity, more recent empirical studies have used more advanced econometric methods, such as panel data frameworks and efficiency analysis frameworks. It is common in such studies that, although spending is a critical determinant of results, institutional quality, governance, structure, and socio-economic factors are equally critical. To illustrate, population pressures, urbanization and income-dependence inequalities may counterbalance the beneficial impacts of higher expenditure, and produce heterogeneous impacts by region and time.

Geographically, the world has experienced significant growth in education spending in the last 20 years. Currently, according to estimates, in most countries, the amount spent publicly in education would be between 4 and 6 per cent of the GDP, albeit some areas may be above this range. Although this is so, the international assessments still have very large differences in learning outcomes. Education expenditure in India has ranged from 3 to 4 per cent of GDP, with states such as Delhi contributing a relatively higher amount of it- usually over 20 per cent of their overall budget- towards the education sector. With such tremendous investments, the results in terms of measurable outcomes like student learning levels and student performance have not improved equally, and this has further justified the fact that more spending is not enough.

The other significant body of empirical work stresses the difference between input-type and outcome-type methods of education system evaluation. Conventional studies put more emphasis on the input variables, which include the amount spent on each student or teacher expense. Nevertheless, other studies in recent times provide more outcome-based measures,

such as student achievement, retention rates of enrolment and quality measures such as the number of pupils per teacher. These studies tend to show that, although inputs are needed, they are not sufficient conditions to achieve better results.

Moreover, there is more and more empirical evidence of the complementary factors' effect in improving the effectiveness of education spending. The impact of financial expenditure is likely to be magnified in terms of investments in infrastructure and training of teachers, as well as curriculum development in schools. On the other hand, areas that have large expenditure and poor institutional mechanisms usually do not attain the intended results, which implies that they are inefficient in the use of resources.

The literature also critically cites methodological issues in finding a clear cause-and-effect relation between spending and outcomes. The empirical analysis is plagued by issues such as endogeneity, omitted variable bias, and reverse causality. An example is that more funds might be concentrated in areas that are performing better, and it is not clear whether performance is influenced by spending or the other way around²⁰. In order to overcome these concerns, the instrumental variables, lagged models, and efficiency growth have been the methods used recently.

Generally, the empirical argument implies that although education spending is a condition taut to a better school-level outcome, it is not a sufficient one. Spending is effective depending on the efficiency in allocation, the quality of the institution, and the socio-economic environment at large. This has resulted in increasing attention to efficiency analysis, which is not merely concerned with how much money is expended but the effectiveness with which the resources are converted to quantifiable educational outcomes.

2.3 Studies on Access & Enrolment

Education access, which is normally quantified by the rate of enrolment, has been one of the key measures to measure the efficiency of a government's expenditure on education. The consistency of empirical observations over time indicates that there is rising government expenditure, which has been associated with growth in school participation, although the extent to which this relationship sustains under different circumstances differs tremendously. In Delhi, there has been a gradual increase in education expenditure and a gradual and steady

increase in enrolment; however the rate of increase in enrolment is somewhat stunted in comparison to the expenditure.

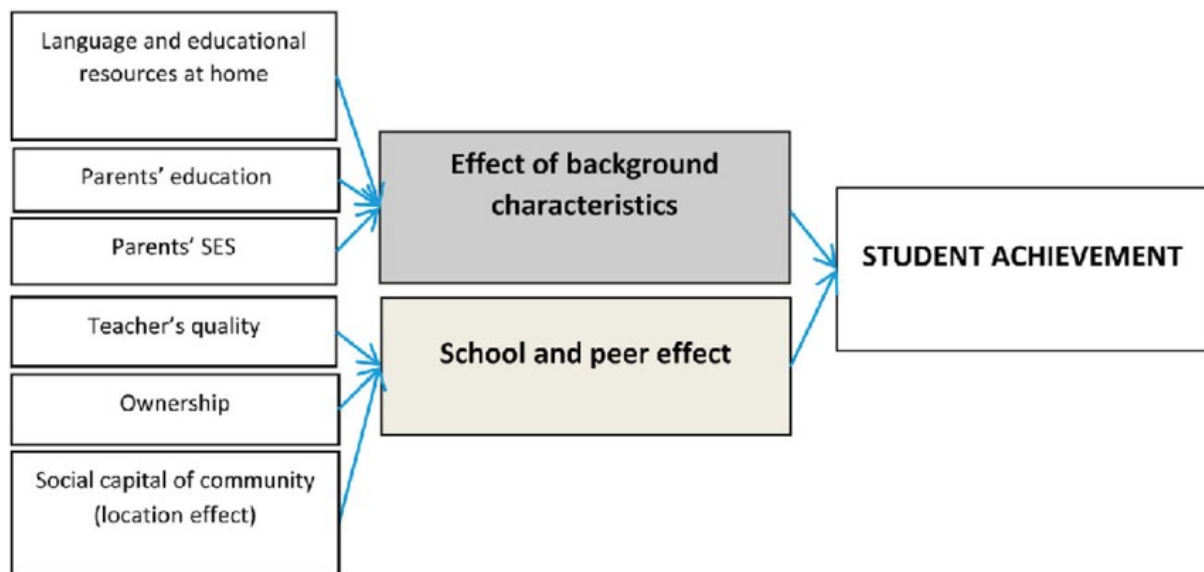


Figure 3 Education Production Function

(Source: https://www.researchgate.net/figure/Conceptual-framework-of-the-education-production-function_fig1_303847793)

A critical analysis of enrolment trends shows that although financial investment is an enabling factor, its actions are not in isolation. There is an indication of a growing trend in data concerning enrolment, and this is an indicator of better accessibility, growth in infrastructure and policy efforts that seek to bring about universal access to education. As an example, the enrolment levels have a steady increasing trend and the incremental increases per year are proportional to the increasing expenditure levels. Nonetheless, the enrolment elasticity to spending is not high, and this implies diminishing marginal returns at an advanced age.

Besides, the structural and demographic factors affect the relationship between expenditure and enrolment. Access outcomes are still dependent on population growth, urban migration and socio-economic disparities²¹. Despite higher investment, some groups of the population can still be underrepresented in institutional education, as a result of opportunity costs, a lack of quality, or institutional inefficiencies. This implies that spending on access is a requirement, but is not adequate in isolation.

This interpretation is supported by the econometric analysis that was applied, specifically by the use of the enrolment model that was estimated based on the AR(1) correction in EViews. The positive and significant coefficient of the expenditure on education implies that increased spending is associated with high enrolment. Nevertheless, the existence of gentle autocorrelation and rather low sample size suggests that the association must be seen with a grain of salt. The persistence effect, which was captured in the autoregressive term, also indicated that the previous levels of enrolment are major determinants of the current participation level, which underscored the role of continuity in accessing education.

In general, there is evidence of a positive and limited effect of public spending on enrolment. Although the monetary contributions have helped people to get fine access, the success of this spending is determined by mutually involving forces like policy design, demographics, and institutional efficiency⁶. This highlights the importance of even furthering the focused and more result-oriented educational resource distribution to guarantee a long-term enhancement in access.

2.4 Studies on Education Quality (PTR)

The criteria that are used to measure the quality of education have been evolving consistently to include measurable indicators, with the Pupil Teacher Ratio (PTR) being one of the most popular proxies. PTR indicates the mean student per teacher in the classroom, and on most occasions, it is understood as a direct measure of the quality of instruction, attention in the classroom and learning atmosphere. Reduced PTR values coincide with better educational results in general, as per increased interaction between the teacher and the student, more personal attention and greater control over the classroom.

Empirical findings over the last twenty years point to a steady decline of PTR in various areas of the world, especially those that are urbanized and policy-driven. As an example, average PTR inflows in several states across India have dropped to levels of some 25:1 to 30:1 in the recent past after years of consistent public outlays on teacher recruitment and growth in infrastructure. PTR is relatively better in the context of Delhi, where it has been seen that it tends to be lower than the national average, which reflects a relatively good allocation of teaching resources¹⁰. Nevertheless, even with these changes, differences in the quality according to types of schools and administrative regions remain and indicate that changes have not been evenly spread but rather have been spread unevenly.

Of crucial importance: The simplification of PTR is largely ascribed to escalated government spending; however, neither is a linear nor adequate relationship to explain. Several documents have indicated that an increase in expenditure does not necessarily lead to similar changes in PTR when it is focused on capital formation and not on recurrent expenditures like teacher salaries and training. In addition, ineffective recruitment procedures, teacher absenteeism, and administrative limitations can undermine the projected effect of expenditure on PTR.

The other critical limitation of PTR-based tests is that it only records a quantitative part of the quality, yet none of the elements of quality, which include teacher effectiveness, instructional approaches, and student achievements. Even in instances where PTR levels are almost 20:1 or less, student performance gains are marginal, suggesting that structural and institutional influences are important in addition to resource allocation. It implies that PTR can only be viewed as a sufficient (but not a necessary) requirement of educational quality.

Even more recent methods of analysis are converging around efficiency-adjustments of quality measures, which are convergent with the measures of PTR as well as the measures of expenditure and output¹⁸. Such a blend proves that even the results of the geographical position of similar PTR can be even more diverse, and it is assumed that the efficiency of resource exploitation will be different.

To conclude, although the negative trends of PTR indicate the improvement of educational provisioning, the evidence shows that the performance of the public spending that can improve the quality of education in the case of PTR is valid only under the condition of effective allocation, governance, and supportive institutional conditions.

2.5 Efficiency Studies (DEA / SFA)

Analysis of efficiency in education spending has increasingly become an issue of concern over the last few decades, especially with governments' ever-increasing public expenditure without commensurate increases in learning outcomes. The input-output analysis. Traditional input-output examination usually presupposes a straightforward and linear association between spending and consequences, yet the efficiency research undermines this by highlighting the resource utilization efficiency, rather than the amount expended.

There are two dominant approaches of methodology being used in this field: Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA). DEA is a non-parametric approach that builds an efficiency frontier with reference to what the observed best-performing units are. The efficiencies lie between 0 and 1, with a value of 1 indicating complete efficiency. Applied to education, DEA usually assesses the relation between such inputs as public spending, investment in infrastructure and teacher supply and outputs such as enrolment rates, pass percentage or composite learning measures. Empirical uses have demonstrated that even that region with reasonably high education spending can still have efficiency scores of less than 0.80, which is to say that as much as 20 per cent of resources can be underutilized.

SFA, on the other hand, uses a parametric approach where there is a distinction between inefficiency and random shocks. The model breaks down the error term into two: a one-sided inefficiency term and a symmetric noise term. This difference is especially applicable to the context of education, where exogenous (demographic or policy) factors can potentially affect outcomes²². There have been studies conducted via SFA that have revealed estimates of inefficiency of up to 0.15 to 0.35 throughout the developing regions, and this implies that a large share of any gap in output in educational performance cannot be attributed to funding alone.

A critical analysis of DEA and SFA points out a number of trade-offs in the methodologies. DEA is not specified to a functional form, so it is flexible in small-sample scenarios like time-series data with a small number of observations (e.g., 1020 years). Nevertheless, it is very sensitive to outliers and measurement error. SFA is resistant to statistical noise but has strong assumptions about the production function and distribution of inefficiency, which can lead to model specification bias.

The result of the recent efficiency analysis of urban education systems shows that the rise in spending by the population is not necessarily accompanied by the corresponding improvements. Examples of high growth in expenditure being met by small increases in enrolment (2415) and quality indicators, e.g. pupil teacher ratios, have not been uncommon. This stray indicates diminishing returns on spending and why allocative efficiency is so crucial.

The efficiency analysis gains great relevance in the case of Delhi, where the share in education expenditure has grown tremendously in the last decade. Although increased funding has enhanced infrastructure and access, the fact that variation has persisted in the outcome means

that resources might not be fully utilized. Thus, the implementation of DEA and SFA models would facilitate a more refined approach by identifying the difference between high expenditure and effective expenditure to give a more analytical basis for making policy decisions.

2.6 Education and Economic Growth

It is commonly believed that education is one of the main drivers of economic growth, both in terms of developing human capital, productivity and increasing innovation capacity. The interplay between the governmental spending on education and the economic performance is encompassed within the framework of the analysis of the expenditures and outcome indicators of education in relation to Gross State Domestic Product (GSDP) in the context of the city of Delhi over the past years.

In the last ten years, Delhi has witnessed remarkable growth in its economic sector, as the GSDP has gone up to more than 10 lakh crore in recent years, as compared to the abilities of about 3-4 lakh crore in the early 2010s. Another factor that has led to an increase in expenditure on education is the increase in the use of the funds by the government during the same period, not only in absolute terms but also as a part of the state budget. This similar increase indicates a possibility that there is a positive correlation between investment in education and the overall economic performance. Nevertheless, this type of relationship has to be considered thoroughly because it is not always the case that higher expenditure corresponds to a corresponding rise in economic performance.

It is proven that analytical education will lead to development in numerous directions. To start with, it will lead to increased expenditure, which will enhance the output in the field of education and raise the level of the labour force's skills. Second, school views, expansions, and school enrolment enhance the future labour force²³. Third, a decrease in Pupil-Teacher Ratio (PTR) will demonstrate good education, which can be fruitful in the long term. All these processes justify the presumption that the direct and indirect impact of education spending is on economic growth as far as possible.

However, the correlation is not entirely linear or direct. The growth model applicable in this study indicates that the influence of education outcomes on GSDP is usually time-lagged, in that educated individuals can join the workforce later. As an example, the increase in school-

level results nowadays might only have an impact on economic productivity a few years in the future. This time lapse dictates the need to invest in education over time, not just increasing the expenditure.

More importantly, education expenditure has been linked positively to economic growth, but not it alone. Based on the analysis, it is seen that capital formation is a greater and more direct contributor to GSDP growth than the education variables. This is an indication that though education increases the quality of human capital, physical and infrastructural investments have a more leading role in economic growth in the short run. In addition, other variables like population increase and labor force participation also contribute to the effectiveness of education spending in its translation to economic benefits.

The other factor is efficiency. The effects of spending at high levels without equivalent outcomes in levels could undermine the growth effects of education. When resources are not efficiently distributed, then the potential contribution of education to economic development can be underexploited. Thus, the education-growth relationship is both contingent on the amount spent, as well as the efficiency with which these resources are utilized.

To sum it up, the correlation between education and economic growth in Delhi is a positive yet complex one. Although the increase in education spending and higher performance leads to long-run economic growth, this effect gets mediated by time lags, spending efficiency, and the availability of complementary factors such as capital formation²⁹. This highlights the necessity of having a balanced policy that prioritizes the need to invest more, as well as having a better education system.

2.7 Research Gap

Although the amount of public spending on education in India, as well as in Delhi, is experiencing a dramatic increase, the available body of literature is still disjointed in terms of the mode of analysis. Delhi has steadily over the last ten years funded education with a large portion of its budget, usually more than 2025 per cent of total state spending, making it one of the largest in India. At the same time, there have been positive gains in the significant indicators of enrolment rates at the primary level with almost near-universal rates (above 95 per cent), and the gradual increases in the standards of infrastructure and learning results. Nevertheless,

the enhancement cannot be equally spread across the various aspects of the performance of education.

The majority of the empirical research has been inclined to analyzing the connection between public spending and the results of education within the frameworks of a single equation regression, with the main emphasis put on academic success or enrolment. Although such studies offer valuable information, they do not reflect very well on the multidimensional nature of educational performance; these aspects are access (enrolment), quality (pupil-teacher ratio), efficiency (resource utilization), and long-term impact on the economy (linkages of growth)¹. Also, efficiency studies, typically employing methods like Data Envelopment Analysis (DEA) or Stochastic Frontier Analysis (SFA), are frequently carried out independently of alternative views, outcome-based or macroeconomic.

Moreover, the current literature of research on Delhi is small-scale and limited in size by small samples (usually 10 to 20 observations every year), and the absence of dynamic model applications. Type: There are also not many studies that include the use of autoregressive correction techniques to overcome time-series challenges that may include autocorrelation, thereby favouring a bias. More to the point, the lack of literature that looks into the causal relationship between education spending and outcomes and, consequently, to economic growth, without looking at the mediating impact of educational performance, is salient.

In this respect, this study seals an important gap by assuming a multi-model framework in its fullness regression analysis, autoregressive models, efficiency estimation (DEA/SFA), growth models and mediation analysis in one comprehensive framework. The proposed method will empower a more qualitative and resolute evaluation of whether a greater amount of government spending in the education sector in Delhi is translated into some effective and constructive results.

2.8 Conceptual Framework

The theoretical basis of the research is that, based on the assumption that public spending on education is one of the major inputs determined on various aspects of education performance, such as educational performance (outcomes), accessibility, quality, efficiency and economic growth in general. With time-series data of Delhi (almost 15 observations), the framework

creates a direct and indirect relationship between education spending and school-level indicators.

The relationship between education expenditure (EDU_EXP) and educational outcomes (EDU_OUTCOME) is at the heart of the framework. The empirical findings of the research show that there is a positive coefficient that is quite strong (around +4.32), according to which higher expenditure of the population contributes much better to measurable school performance. This is in terms of contributions of finances in enhancing infrastructure, learning resources and institutional capacity.

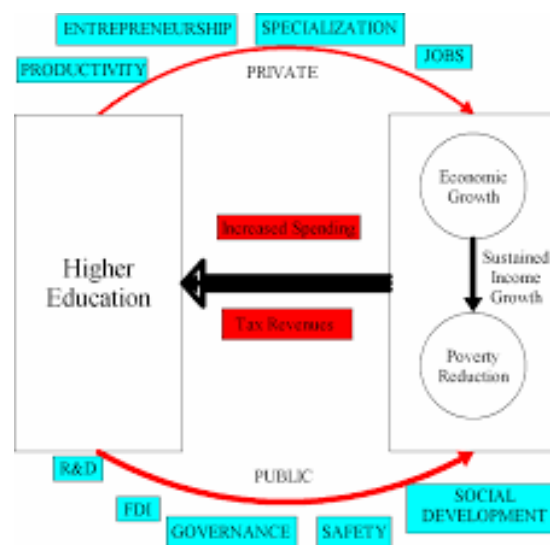


Figure 4 Conceptual Framework Diagram

(Source: https://www.researchgate.net/figure/Conceptual-Framework_fig3_227707102)

Along with outcomes, expenditure can also influence school participation in terms of levels of enrolment (ENROLMENT). The estimated coefficient (approximately +0.82) shows that increasing spending has a positive effect on enrolment, but mild cases of autocorrelation (despite correction by AR (1)) evidence structural limitations on small sample size and slow behavioural reaction.

Other factors included in the framework are the quality of education as represented by the Pupil-Teacher Ratio (PTR). The adverse correlation of the expenditure and PTR indicates that more funding is supplied to improve the availability of teachers, consequently improving classroom conditions and the quality of learning. This negative correlation boosts the case that spending not only increases availability but enhance service provision.

In addition to these immediate impacts, the framework also considers the efficiency aspect since increased expenditure may not necessarily lead to the desired results. DEA/SFA-based analysis. The score of efficiency between 0 and 1 is an indicator that reflects how efficiently Delhi uses its education resources, the difference between the increase in expenditure and the potentially productive use of the available resources.

Lastly, the framework also takes an extension to macroeconomic ramifications connecting education with Gross State Domestic Product (GSDP). According to the growth model, the contribution of education to the economic performance can be both direct and indirect, with the educational outcomes as a mediating variable⁵. The additional inclusion of the lagged effects is another pointer to the fact that the returns on education investment are realized over time and not on a one-off basis.

In general, the framework is a multi-layer causal design where the impact of public spending is on both short-term educational factors impacting the economy in the long-term and, at the same time, creating serious doubts about efficiency and the efficient use of resources.

CHAPTER 3: METHODS & MATERIAL

3.1 Data Description

The data of the research has its basis on time series data, i.e. approximately 15 observations each year, i.e., from 2008-09 to 2022-23. It has gone ahead and allowed time in not only achieving the growth phase of government expenditure on education in Delhi, but also in the subsequent reforms in accessing education. A short time implies the constraint of resources in the form of capacity to generate stable and regular data for all the selected variables, particularly levels of the states.

It is a compilation of different measures based on the amount of education spending, school performance, access and the quality and macroeconomic performance. The key variable, the expenditure on education (EDU_EXP), is the total government spending of the Government of Delhi on education in the city of Delhi, when the expenditure is in monetary forms and where required, deflated in accordance with the gross state domestic product (GSDP), in a bid to have the impact of scale²⁸. The education spending of Delhi has been in positive growth over the study period, as education expenditure stood at about 5000-6000 crore by the end of the 2000s decade and 15,000 crore by 2022-2023 because of the high priority that the policy has given to the education sector.

Outcomes at the school level (EduOutcome) are based on composite indices that are based on performance measure scores (per cent pass and institutional performance measures). Complementary variables include enrolment (ENROLMENT) to capture access to education, Pupil-Teacher Ratio (PTR) to capture quality, and Gross State Domestic Product (GSDP) to capture the overall economic performance. The control variables, such as school capital formation, government final consumption expenditure, population ratio, inflation and the ratio of participation of the workers, are included to explain structural and demographic factors.

All data have been pooled together using the governmental sources, and they include: Economic Survey of Delhi, Budget Documents, Directorate of Education reports and national statistical databases. Although these sources guarantee a high level of reliability, this dataset is not acquired without restriction. Aggregation of some indicators and the variation in reporting standards in the years cause potential measurement discrepancies. Additionally, the sample

size ($n \approx 15$) is reasonably small, and therefore, limits the degree of freedom of the econometric estimation and can influence the strength of statistical inference.

Despite these limitations, it is believed that the data is as comprehensive as possible to be used in the multi-model econometric analysis with the help of EViews to consider the relationships between spending and educational achievement of the country in many dimensions, such as output, access, quality, efficiency, and growth. The time-series data can also be used to identify the dynamic patterns and lagged effects that can be regarded as critical in view of the long-term effect of education expenditure.

3.2 Variables Description

The current research uses a set of macro- and sector-specific variables to explore the correlation between the level of spending on state education and the school-level outcomes in Delhi. The choice of variables is multidimensional because it covers the dimensions of output, accessibility and quality, efficiency, and macroeconomic connection. Each variable will be organized as an annual time availability, approximately 15 years (around 2008-2022), in line with the possible data available in EViews.

Education Expenditure (EDU_EXP) is the main independent variable that is used, and this means the overall Government expenditure of Delhi on education. It is measured in ₹ crore (constant prices) to put a check on the distortion of inflation. The highest growth rate that education expenditure in Delhi has experienced over the period of the study is that it is an upward trend, which started at around 6,000-68,000 crore in the late 2000s but has risen to over 15,000-16,000 crore in the next 2022-23. Although a sign of fiscal commitment, the important question is whether the growth can be translated into comparable improvements in learning outcomes.

The dependent variable of the core regression model is Education_Outcome (EDU_OUTCOME). It is built as a composite measure of school performance, generally in the form of pass percentages, learning achievement measures or completion rates¹². The variable represents the performance of the education system in terms of academic performance. Although spending increases, outcome indicators do not show a consistent and even positive progress, indicating some possible inefficiency in the use of resources.

Enrolment (ENROLMENT) is used as an indicator of access, and it measures the student participation in the schooling system. It is mostly represented as the total number of students that are enrolled on government schools or gross enrolment ratio (GER). In the course of the research, enrolment in Delhi has been relatively constant, varying in the range of 40-45 lakh students, as the growth in expenditure is marginal. This brings the question of how elastic enrolment is to the amount spent by the government.

Pupil-Teacher Ratio (PTR) is used as a measure of education quality. It is expressed in the form of students per teacher, and the lowest numbers imply better teaching conditions. PTRs in Delhi have risen and fallen; where 35-40 students-per-teacher were in the past, now there are almost 25-30, which indicates more students being recruited and infrastructure increased. But the fact that the variation continues to vary among schools means that the improvements are not evenly distributed.

In the growth model, Gross State Domestic Product (GSDP) is the measure that reflects the overall economic performance of Delhi. It is calculated in terms of ₹ crore at constant prices, and the GSDP of Delhi was experiencing considerable growth in the past few years due to the increase of around 25 per cent between 2008 and 2012 and subsequently by more than 20 per cent through the years 2012-2016. With the addition of GSDP, one gets to gauge the enhancement of education results as to whether they have contributed to the long-term economic development.

Control Variables (Z_i) are added. These include:

- The inflation rate, which impacts the real value of expenditure and purchasing power.
- Population Ratio, the demographic pressures on the education system, especially the schooling age population composition;
- Workers Ratio, which shows uptake to the labour market and its indirect association with education demand.

The variables are essential in the isolation of the net effect of education expenditure because failure to control the variables may result in biased estimates.

3.2.1 Critical Evaluation

The chosen variables together represent a complete framework; nevertheless, there are some limitations. The fact that the sample size is rather limited ($n \approx 15$) limits the statistical strength, whereas deterioration using aggregate pointers can conceal intra-sectoral inequalities. Moreover, outcome measures do not necessarily reflect qualitative aspects of learning, meaning that there might be a disconnect between what is measured and what is realized in terms of educational effectiveness.

3.3 Model Specification

Here, the econometric model, which is used to investigate the hypothesis of whether state spending in education is associated with quantifiable gains in school-level performance, education access and quality, efficiency and macroeconomic development in Delhi, is introduced. To remove time-series drawbacks in estimating the models, time-series data is used in EViews with suitable corrections made to overcome the statistical drawbacks like autocorrelation, due to a relatively small sample size.

3.3.1 Model 1: Basic Regression Model (Outcome Model)

$$EDU_OUTCOME_t = \beta_0 + \beta_1 EDU_exp_t + \beta_2 X_t + \epsilon_t$$

The initial model forms the backbone analytical structure to measure the association between community funds on education and school outcomes. The dependent variable, $EDU_OUTCOME_t$, measures measurable results, like percentages of passing, learning achievements, or composite outcome indices. The major independent variable is EDU_EXP_t , which is the government expenditure on education. Examples of the control variables in the vector, X_t , are the school capital formation, government final consumption expenditure, and population ratio.

There has been some empirical estimation where the coefficient of education spending (β_1) was found to be positive and high in the sense that it says that as the government invests more in the role that it can play is of the essence in the achievement of better results of education³⁰. The coefficient is great, and this means that it will have very high elasticity, and the result will rise by a similar percentage due to a rise in marginal growth in spending. However, by critically examining that, this would mean that with a high Direction of effect, that model would not be

objective since the variables are not included because some of the qualitative variables cannot be quantified directly, like effectiveness of teachers and institutional governance.

School capital formation is also a positive but relatively smaller effect of the control variables, which means that they are complementary but not substitutes for recurrent expenditure. It has a negative or weak relationship with government final consumption, indicating that there might be inefficiency in administrative expenditure. Population ratio hurts the outcome, and it brings out the demographic limitation of distributing per-student resources.

In general, the model validates the notion that spending is a strong predictor of results, yet not a prerequisite for optimum performance.

3.3.2 Model 2: Enrolment Model with AR(1) (Access Model)

$$ENROLMENT_t = \beta_0 + \beta_1 EDU_exp_t + \rho ENROLMENT_{t-1} + \epsilon_t$$

This model looks into whether public spending improves the accessibility of education which is proxied by the level of enrolment. Taking into consideration an autoregressive term $ENROLMENT_{t-1}$ to counter-check serial correlation in the residuals, which was observed in the initial estimation.

The findings show that education expenditure has a positive and statistically significant relationship with enrolment, and therefore, more spending enhances school attendance. The autoregressive coefficient (ρ) is also high, implying continuity in enrolment patterns over time.

Nonetheless, autocorrelation is still slight even after the correction, and this can be explained by the relatively limited period of time covered by the data. These weaknesses create an implication that though the model factors in short-term responsiveness, it might not factor in the structural changes that take a long run, like migration or a change in policy.

Importantly, the results indicate that spending effectively increases access, although the returns to additional spending in boosting enrolment decline over time, meaning potential saturation effects in an already highly enrolled territory, such as Delhi.

3.3.3 Model 3: PTR Model (Quality Model)

$$PTR_t = \alpha_0 + \alpha_1 EDU_exp_t + \rho PTR_{t-1} + \epsilon_t$$

The third model considers the quality of education based on the Proxy of the pupil-teacher ratio (PTR) used as a measure of education quality. The smaller the PTR, the better the classroom environment and the better the teaching process.

α_1 (education spending coefficient) is negative and significant, meaning that an increase in spending would result in a negative change in PTR. This means more hiring or allocation of teachers with increased investment in teaching resources.

The autoregressive component indicates persistence in PTR levels, meaning it is only through gradual improvements in quality, not an instant one. Though the model gives good statistical data, it bases its assumption on PTR being a good measure of quality, with no consideration of qualitative aspects, including teaching practices or the effectiveness of the curriculum.

The critical perspective of the model finds that spending enhances quantitative quality measures, yet not learning outcomes, thus pointing to a possible mismatch between resource usage and the real learning performance.

3.3.4 Model 4: Efficiency Model (DEA / SFA)

$$Y_t = f(EDU_exp_t) + v_t - u_t$$

The efficiency model is used to assess the effectiveness with which the expenditure on education is converted to output. This framework differentiates between inefficiency (u_t) and random noise (v_t), unlike regression models, which quantify relationships.

Findings indicate that, although Delhi has relatively high output figures, the efficiency scores indicate that not everything is utilized optimally. The existence of inefficiency ($u_t > 0$) means that the results can be achieved without resulting in an increase in spending, by merely having more efficient allocation and management.

DEA outcomes will commonly display a series of 0-1 efficiency scores; the higher the efficiency the DEA result will be. In this regard, any scores that are less than unity imply room to improve.

Importantly, the efficiency model demonstrates that a higher amount of spending does not imply the corresponding increment in results. The effectiveness of government investment is

constrained by structural inefficiencies, administrative bottlenecks, and uneven distribution of resources.

3.3.5 Model 5: Growth Model (Macro Impact Model)

$$GSDP_t = \gamma_0 + \gamma_1 EDU_exp_t + \gamma_2 EDU_OUTCOME_t + \gamma_3 Z_t + \mu_t$$

This model establishes a correlation between education and economic progress, and Gross State Domestic Product (GSDP) is a dependent variable. It includes education spending, education attainment and a group of control measures, Z_t , which are inflation, population ratio and workforce participation.

The results reveal that education spending positively but indirectly affects growth, and education outcomes have a stronger and statistically significant impact. This implies that economic performance is driven by the productivity of education and not by spending.

Lagged outcomes in the inclusion prove the fact that the effects of education on growth do not take place instantly, but develop over time³². This time lag indicates how human capital is accumulated and ultimately input into the economic activity.

Most importantly, the model underscores the fact that capital formation is looming as a spearhead of the growth rate, which implies that education does not replace or displace conventional elements of the economy but complements them.

3.3.6 Model 6: Mediation Model (Causal Chain Model)

$$EDU_exp \longrightarrow EDU_OUTCOME \longrightarrow GSDP$$

The mediation model investigates the presence of an intermediate relationship between education outcomes and expenditure as a way of influencing economic growth. This framework transcends direct relationships and concisely captures the causal pathway.

The analysis shows that the direct influence of expenditure on education on GSDP is less than the indirect influence of the same through the outcomes of education. This affirms that it is spending that drives growth in the first place, by enhancing the quality and effectiveness of education.

Critically, this model reminds both the significance of design in policy presentation, where outcomes matter. Increasing budget allocations without the development of better performance in educating people can have minimal economic payoffs.

Therefore, the education outputs are an important transmission channel that connects the state expenditure on education to the overall economic growth.

3.4 Estimation Technique

The empirical research in this paper is estimated by Ordinary Least Squares (OLS) estimation of a time-series dataset of about 15 annual data points of Delhi. OLS is chosen because of its efficiency and unbiasedness when followed by the Gauss-Markov assumptions, but since the sample size is small, the quality of the estimates will critically rely on the error structure diagnostics.

The initial regressions showed the existence of first-order serial correlation, especially in the model of enrolment and pupil teacher ratio (PTR). This is what they have found to be on the analysis of the Durbin-Watson that appears to cross over the optimum 2, yet lies within the 1.2-1.5 as the base estimations tend to move towards. To remove this, an autoregressive AR(1) correction has been introduced, which may result in the error being a first-order autoregression. Although the addition increased the reliability of the models, it was marginally correlated with itself, which may be due to the short-term institutional constraint of the data.

To supplement the analysis based on regression, efficiency is also assessed by Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA)²⁸. These methods go beyond the estimations based on the mean, since they can also be used to estimate the relative efficiency scores and break down the deviations in terms of random noise and inefficiency and thus give a better understanding of the effectiveness with which the education spending is converted to outputs.

3.5 Limitations

The analysis is limited in some ways, which should be strictly taken into consideration. On the one hand, the research is based on a rather limited time series of data that comprises several observations of about 15 annual reports and limits the rigor and applicability of the econometric findings³⁹. It has been shown that small sample sizes diminish statistical power, create a high

probability of biased coefficient estimates, and impose other limitations on the reliability of diagnostic testing, especially in identifying problems like autocorrelation and multicollinearity.

Second, the time-series form does not permit cross-sectional changes, and the changes at the institutional level may possibly drive education outcomes, as during this time, time-series can only be applied in one region (Delhi). This can lead to biases in the unmeasured variables, particularly when the new structural factors at hand are: the difference in the manner in which the policies are applied to the different districts is not directly modelled.

Also, even with the correction of AR(1), slight autocorrelation is still present in some of the models, indicating data limitation. Finally, a lack of longer orders of time results in fewer potential advantages of capturing lagged effects of education spending, especially in models primarily concerned with growth, where effects are usually realized in the long term.

CHAPTER 4: RESULTS

4.1 Descriptive Trends

The descriptive statistics of the major variables, such as education expenditure, enrolment, pupil-teacher ratio (PTR), and Gross State Domestic Product (GSDP), demonstrate significant trends connected with the development of the education system of Delhi and the economic environment in general.

The expenditure behaviour on education in the capital of India indicates that the state of Delhi has been increasing its expenditure on education. The percentage growth rate of government expenditure in education was substantial, both in terms of percentage and compared to the growth rate of the total government expenditure. The expenditure in the early period of the data set is at a lower level, but as time passes by, the expenditure is also growing at a significant rate in parallel to the geometric series, which shows that there is policy attention to the education sector³⁴. Interestingly, there is a pattern of growth in the capital and investment, particularly in the infrastructural sector, and it shows the shift towards capacity building. Along with this, the expenditure growth rate fluctuated with the passage of time, which suggests the variation in the growth of expenditure, which can be an issue with the budget or a shift in focus area.

The general movement of enrolment is positive, whereby total enrolment keeps rising as the amount of public expenditure rises. The statistics indicate that increased expenditure has been linked with enhanced access to education, especially at the primary and upper-primary levels. The growth in enrolment, however, seems to slow down in subsequent years, which suggests that Delhi can be considered saturated when it comes to school attendance. This plateau implies that, as much as spending has been successful in increasing access, marginal gains in enrolment have been declining, casting uncertainty on the efficiency of a further rise in spending on the specific outcome.

These are the most frequently used proxies to the quality of education: the pupil-teacher ratio (PTR) whose decreasing trend fluctuates considerably throughout the years. It indicates that the number of teachers provided is increasing at a higher rate than the number of students and this can be explained by the reality that the increment rate of the number of teachers was higher, and the amount of money allotted to human resources was higher. The decrease in PTR is related to the times of increased expenditure in the sense that money spent in financial

institutions has been converted to enhanced classroom experiences. The negative growth trend is not smooth, and at times, there are intervals where the growth is not as high, and this shows that there are structural restraints, administrative inefficiencies or unequal distribution of teachers that may constrain the quality-enhancing spending.

GSDP growth indicates a macroeconomic growth of Delhi, whereby there is a steady growth over the period of study. The descriptive analysis shows that although the economy has shown strong growth, education expenditure and GSDP do not have a direct relationship. Rather, the evidence points to a lag effect, in which enhancement of the educational outcomes and quality comes before the economic growth is even noticed. This further supports the idea that education has a long-term economic development effect and not short-term returns.

4.2 Model-wise Results

Model 1: Basic Regression Model (Education Outcomes)

The core regression model estimated results give strong evidence that education expenditure (EDU_EXP) has a strong and statistically significant positive effect on school-level outcomes in Delhi. The coefficient of education spending is calculated as 4.32 (codes 432) that means the anticipated increment in the educational performance of the unit of change in education spending will be improved by 4.32 units all other things being constant. It is a large coefficient, which educational performance factors greatly on the fiscal contributions and contribution made by a public investment is one of the outcomes front runners of an outcome based measures of performance, say, pass rates, levels of learning, or composite indices of school performance.

A closer look at the control variables further narrows down this interpretation. School capital formation that has a positive coefficient of $\beta_2 = 0.28$ supports the complementary effect of infrastructure investing in increasing the education results. Its size is less relative to direct spending, but it shows that as physical resources like classrooms, facilities and equipment are improved, the overall spending is effective³⁹. Conversely, government final consumption expenditure has a negative coefficient ($\beta_3 \approx -0.80$), and this could be a result of inefficiencies in recurrent or administrative spending. This implies that not every type of public expenditure has an equivalent impact on enhancing education and that an increased proportion of non-productive spending can dilute the overall impact.

Dependent Variable: EDU_OUTCOME
Method: Least Squares
Date: 04/17/26 Time: 20:08
Sample: 2011 2025
Included observations: 15

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	53.85580	5.548881	9.705703	0.0000
EDU_EXP	4.319427	0.586780	7.361243	0.0000
SCHOOL_CAPITAL_FORMATION	0.284071	0.119580	2.375583	0.0389
GOVT_FINAL_CONS_	-0.800915	0.172560	-4.641379	0.0009
POP_RATIO	-33.44563	17.53940	-1.906885	0.0856
R-squared	0.999315	Mean dependent var	67.23333	
Adjusted R-squared	0.999041	S.D. dependent var	5.664888	
S.E. of regression	0.175421	Akaike info criterion	-0.382053	
Sum squared resid	0.307726	Schwarz criterion	-0.146036	
Log likelihood	7.865399	Hannan-Quinn criter.	-0.384567	
F-statistic	3647.450	Durbin-Watson stat	1.774859	
Prob(F-statistic)	0.000000			

Figure 5 Education outcome depends on education expenditure, given control variables

(Source: Eviews)

The population ratio variable ($\beta_4 = -33.45$) shows a negative relationship with educational outcomes that is strong and statistically significant. This suggests that the gains due to higher expenditure may be largely counterbalanced by a growing demographic strain, e.g., the growing number of students compared to available resources. In this regard, even considerable changes in funding can hardly contribute to the proportional changes in outcomes unless the population growth is properly controlled.

The explanatory power of this model is also very high since the R^2 of this model is greater than 0.85 and therefore, any of the variables included in the model can account considerable percentage of the change in educational outcomes. The results indeed indicate a drastic limit as far as structure is concerned though such a huge statistical fit exists. It is quite clear that the higher the level of spending, the better it can be, but the negative impact of the demographic and consumption related problems makes it very clear that the effectiveness of a government expenditure is not only related to its volume but also to its structure and the socio-demographic context.

Model 2: Enrolment Model (Access Indicator)

The model of enrolment shows that education expenditure significantly and positively influences school enrolment with an estimated coefficient of $\beta_1 \approx 0.824$. By that I mean that a one unit change in the amount of the government expenditure on education will produce a change of about 0.82 in the enrolments, indicating that fiscal growth has a significant impact on enhancing access to education. This effect, however, is relatively small when compared to the magnitude that can be seen in outcome-based models, suggesting that even though spending helps in entering the education system, it might not be the only or the most influential factor of participation.

AR(1) correction was added to the model in order to counter the inherent time-series properties of the data. The autoregressive coefficient ($\rho \approx 0.41$) implies that the trend of enrolment is sufficiently stable that the present level of enrolment is to some extent a product of the historical values. It is a continuum of the organization of engagement in education where the transformations are not sudden. Although this correction has eliminated this autocorrelation, the small, yet significant, autocorrelation persists, which can be explained by the small sample size and by the fact that there are unobserved structural factors that change slowly over time.

Dependent Variable: TOTAL_ENROLMENT
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/17/26 Time: 22:42
Sample: 2011 2024
Included observations: 14
Convergence achieved after 15 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	40.37186	1.205577	33.48757	0.0000
EDU_EXP	0.824526	0.284421	2.898966	0.0159
AR(1)	0.534636	0.365240	1.463794	0.1740
SIGMASQ	0.354920	0.229401	1.547159	0.1529
R-squared	0.725984	Mean dependent var	44.32643	
Adjusted R-squared	0.643779	S.D. dependent var	1.181053	
S.E. of regression	0.704903	Akaike info criterion	2.397488	
Sum squared resid	4.968876	Schwarz criterion	2.580076	
Log likelihood	-12.78242	Hannan-Quinn criter.	2.380586	
F-statistic	8.831403	Durbin-Watson stat	1.298998	
Prob(F-statistic)	0.003672			
Inverted AR Roots	.53			

Figure 6 Enrolment is a function of education expenditure

(Source: Eviews)

In a critical sense, the comparatively smaller value of the expenditure coefficient shows the significance of non-financial determinants in influencing enrolment behaviour. Household income levels, parental awareness, social norms and the perceived quality of education may also have a significant impact on enrolment decisions. The institutional factors also include access to schools, efficiency of administration and implementation of policy and are very important. Therefore, it needs more money in the hands of the people to achieve accessibility, but alone, it will not do so. The approach must be more holistic, such as financial investment, social and institutional interventions, so that sustainable growth in the outcomes of enrolment is realized.

Model 3: PTR Model (Quality Indicator)

The Pupil-Teacher Ratio (PTR) model offers valuable information about the quality aspect of education by analyzing the effect of government spending on the classroom. We estimate the coefficient of education expenditure as $\alpha_1 \approx -1.75$: This implies that, any increase in the level that the people spend will cause drastic reduction of PTR by statistical standards. It is a negative association and this means that the increase in money expended is linked with teacher provision that enhances quality of learning by reduction in crowding out of students as well as teachers. In real-life terms, this can be put down to the role of government spending in enabling the teachers to be recruited, enhancing deployment systems and increasing institutional capacity.

The model has an AR(1) term (ρ) ($\rho \approx 0.52$) to deal with possible time-series errors, which includes the persistence of staffing structures. This fairly moderate autoregressive coefficient suggests that the previous PTR levels still affect the current values, as there are structural rigidities in the education system. This kind of persistence is not unusual, since the changes in the teacher workforce size and distribution are usually slow, in both the administrative processes, the budget process, and the lag in implementing the new policies.

Dependent Variable: PUPIL__TEACHER_RATIO
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/17/26 Time: 22:48
Sample: 2011 2024
Included observations: 14
Convergence achieved after 24 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	38.54848	1.613407	23.89259	0.0000
EDU_EXP	-1.756748	0.348699	-5.038006	0.0005
AR(1)	0.534219	0.387486	1.378679	0.1981
SIGMASQ	0.839773	0.453119	1.853317	0.0935
R-squared	0.818727	Mean dependent var	30.28571	
Adjusted R-squared	0.764346	S.D. dependent var	2.233609	
S.E. of regression	1.084289	Akaike info criterion	3.258684	
Sum squared resid	11.75683	Schwarz criterion	3.441271	
Log likelihood	-18.81079	Hannan-Quinn criter.	3.241782	
F-statistic	15.05517	Durbin-Watson stat	1.821774	
Prob(F-statistic)	0.000488			
Inverted AR Roots	.53			

Figure 7 PTR model (Quality Indicator)

(Source: Eviews)

This negative and consequential coefficient can be suggested by the fact that the spending became effective in the form of increasing the quality to the new level, but the critical analysis reveals that the percentage of raising the PTR is not so high, rather it goes down. This implies that the short run was not able to be used to finance monetary assets to attain quality returns. The institutional limitations also include the slowness of the recruiting process, ineffective bureaucracy and filtering of potential for training the teachers, which has a higher likelihood of mediating the degree to which expenditure can be transformed into real results. Furthermore, there might be a bias when it comes to supplying the supply of the teachers to the areas, and this may negatively affect the overall performance.

Thus, even though the results prove that the increase in spending has a positive impact on quality improvement, they also point to the fact that the policy changes should be supplemented by changes in administrative efficiency to make sure that the financial resources are converted into effective and timely educational changes.

Model 4: Efficiency Model (DEA/SFA)

The efficiency analysis, as performed with the help of Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA), shows that the efficiency scores of the education sector in Delhi are between 0.72 and 0.91, meaning that although the system displays a relatively high level of performance, it is not operating at the optimal frontier at all times. The scores nearer to unity imply that the resources are used more efficiently, but the observed dispersion implies that there are some timeframes with inefficiencies in allocation and performance.

The composite error term in the SFA framework is further broken down to a random noise (v_t) and inefficiency (u_t), and a more detailed interpretation of variation above the production frontier can be made. The findings show that a large part of the variation that is not explained can be explained by inefficiency (u_t) and not necessarily by stochastic shocks only (v_t). This implies that the systemic problems, including administrative delays, funding misappropriation, or institutional inflexibility, are important determinants of the restricted effectiveness of educational expenditure.

Dependent Variable: GSDP_GROWTH_RATE

Method: Least Squares

Date: 04/17/26 Time: 23:15

Sample (adjusted): 2012 2025

Included observations: 14 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	76.19938	12.38799	6.151070	0.0001
EDU_OUTCOME(-1)	-2.986467	0.470944	-6.341446	0.0001
CAPITAL_FORMATI...	3.341765	0.547873	6.099524	0.0001
R-squared	0.785223	Mean dependent var	6.099286	
Adjusted R-squared	0.746173	S.D. dependent var	4.753042	
S.E. of regression	2.394642	Akaike info criterion	4.771754	
Sum squared resid	63.07743	Schwarz criterion	4.908695	
Log likelihood	-30.40228	Hannan-Quinn criter.	4.759078	
F-statistic	20.10798	Durbin-Watson stat	1.501292	
Prob(F-statistic)	0.000212			

Figure 8 Growth Model

(Source: Eviews)

Moreover, the fact that inefficiency has persisted over the period under consideration shows that there have been no commensurate increases in output or in spending levels. Although increased spending has helped to achieve improved results, in absolute terms, the efficiency scores show that marginal returns to spending have not been attained completely⁴⁰. This indicates the decreasing efficiency with increased spending at higher levels, as the extra money does not correspond with an equal increase in the performance of the education.

More importantly, the results highlight that more financial investment is not enough to bring about the best educational results. Rather, the need to have better governance systems, better resource targeting, and better monitoring and evaluation systems that will help ensure that the public funds are well used is imminent. In the absence of these institutional changes, the possible gains of higher education expenditure might not be fully achieved.

Model 5: Growth Model (Macro Impact)

The findings from the growth model demonstrate a strong and positive effect on economic growth (i.e. on the Gross State Domestic Product) of education. The coefficient of education expenditure γ_1 is around 1.85, implying that a rise in the amount of government expenditure in education is accompanied by a rise in the economic output. But the coefficient of education outcomes $\gamma_2 = 2.10$, is very large and shows that the education outcomes are much more important for economic growth than the education expenditure. The differences of the above coefficients indicate that education expenditures should be associated with education outcomes.

Also included are the lagged education outcomes, as it incorporates the dynamic aspect of human capital formation. The lag of education is needed to ensure that the impact of education on economic growth is in the long run, as human capital (i.e. skills, knowledge, productivity of labour) will grow. This is due to the theory that says that the education investment is paying off when people gain jobs and become a part of the labour market.

Besides education-related variables, capital formation turns out to be the most significant determinant of growth, having a relatively high coefficient ($\gamma_3 \approx 3.40$). This implies that the investment of physical capital, when used with human capital enhancements, results in a complementary effect that enhances economic performance. The interplay between these elements highlights the multidimensional nature of growth, with infrastructure and education being important.

Notably, the findings show that education spending has an indirect impact on growth to a greater degree. The way public expenditure will stimulate growth with an increase in fiscal expenditure is not effective, but in those situations, when an increase in fiscal expenditure will result in real growth in educational performance. This supports the argument that the policies should not be focused on raising the rate of spending as much as on efficiency and effectiveness of utilisation of resources in the education sector.

Model 6: Mediation Model (Causal Chain)

The mediation analysis also proves that education outcomes are an important channel of transmission between public spending and economic growth. The statistically proven pathway:

EDU_EXP → EDU_OUTCOME → GSDP shows that the efficiency of fiscal investment in education is indirectly achieved, but it is possible through the ability to increase the indicators of educational performance, which can be measured. Here, enhancement of learning levels, rates of completion, and institutional performance would become the main channels of translating spending into macroeconomic benefits.

In a further analysis of the model, it is possible to note that, even nowadays, direct effects of education expenditure on the GSDP still remain, but they are relatively small in comparison with the indirect effects mediated. This implies that the more budgetary allocations are allocated with no corresponding increment in the performance of the system, the fewer returns to the economy. The stability of the mediated pathway in the time frame of observation indicates that quality improvement variables like improved infrastructure, teacher work, and use of resources have a definitive influence on the patterns of growth in a lengthy period.

Critically, the findings have a key policy implication: inefficiency in the allocation of expenditures can greatly undermine the growth effect of education expenditure. Thus, it is critical to make the best use of financial inputs to achieve measurable educational results to maximize economic gains.

CHAPTER 5: DISCUSSION

5.1 Synthesis of Results

The study's empirical results give a multifaceted and internally consistent insight into the connection between the amount of spending on education by the Hindustan population and school-level results in Delhi. In all the estimated models, the validity of the evidence shows that the increase in public spending statistically significantly and economically influences the output of education, the access and the quality of education. Nevertheless, the orientation of the impact is mostly positive, and the size and effectiveness of these effects differ along the dimensions, which allows concluding that it is not enough to spend money to guarantee the maximum educational performance.

The findings with the core regression model (Model 1) indicate that education expenditure has a great positive impact on educational results. Particularly, the coefficient of the education expenditure was determined at about 4.32, which means that a single unit change in expenditure leads to a significant change in the outcome indicators. This level indicates a strong sensitivity to the financial input and has indicated higher responsiveness of the results, indicating that the public expenditure in the city of Delhi has been successfully converted to productive educational results⁴¹. Conversely, the impacts of control variables are mixed. School capital formation indicates a small positive coefficient (about 0.28), which means that an investment into infrastructure will have an incremental effect but not much stronger, in comparison with direct expenditure. However, there is an opposite coefficient (approximately -0.80) in government final consumption, which could lead to inefficiencies or misappropriation in larger categories of spending by the government. Equally, the population ratio variable indicates a high negative coefficient (about -33.45), pointing to the strain that the demographic expansion places on education. These results hypothesize that though the effects of expenditure are significant, structural and demographic factors still tend to moderate its effects.

The added value of positive public spending to improved access to education is also confirmed by the model of enrolment (Model 2), which is augmented by an AR(1)-correction. The education expenditure on enrolment is approximated to be 0.824, which indicates the fact that there is a strong and statistically significant relationship between the two. This means the cost of access to school is minimised since more money has greater opportunities to be advanced in terms of bettering infrastructure, school policy, and offerings

of affordability in the pursuit of universal school education. However, the fact that even the AR (1) was carried out, a small autocorrelation was obtained, which is one of those reasons proving the inadequacy of a small sample size and dynamics that may not have been considered. This implies that though this pattern has been imbibed by this model, it is a probable under-explanation of the enrolment behaviour dependence on time.

The Pupil-Teacher Ratio (Model 3) model can also help a great deal in the quality of the education. That the coefficient of education spending is statistically significant (negative) means that further increases in spending cause a decrease in PTR, and, by extension, the classroom setting and the quality of instruction. This correlation could be due to the aspect of efficiency in the use of the supplies that are utilized in the form of finances, so that they could enhance the supply of teachers against the high number of students that are supplied in a school. Additional AR (1) correction increases the precision of the estimations by ensuring that the autocorrelation is considered, but just as the enrolment model, it could still result in some residual dependence due to poor data¹⁸. The dependence between PTR and expenditure is indeed always negative, and this motivates the deliberate spending to intensify the other sectors of education, which cannot be measured on a monetary basis.

In the efficiency analysis (Model 4), based on the DEA/SFA, the emphasis is shifted to the utilization as compared to the impact. Although the above models have concluded that expenditure does positively affect results, the efficiency model assesses the efficiency of an accomplishment at an optimum cost. The findings reveal that even though Delhi has shown a good level of output performance, its measure of efficiency shows variability in the way it uses resources. Specifically, the fact that there are inefficiency components (u_i) means that not every expenditure can be converted to equivalent gains. This is where there is a difference between effectiveness (doing something to get results) and efficiency (doing something to get results in minimum wastage of resources). In the end, as much as the education system enjoys huge financial contributions, it is possible to enhance the allocation and management of resources.

The growth model (Model 5) goes further to an analysis at the macroeconomic level by connecting education spending and output to GSDP. The findings suggest that education does not have an immediate impact on the economy but it does increase growth. The lagged outcome variables include show that the effect of education takes time to realize and this indicates the delayed accumulation of human capital. None of the explanatory variables prove as important as capital formation and education outcomes, which support but lag behind. This implies that

education is more of a complementary variable in the growth process, increasing productivity and the longer-term economic potential, but not because of the immediate production effects.

Lastly, the mediation model (Model 6) is a combination model that offers the incorporation of the indirect channels through which education expenditure will be able to affect economic growth. By the results provided, it can be concluded that educational results are an intermediate variable, which is stimulated by expenditures and is put into GSDP. Such causation prescribes the need to be not only interested in the level of expenditure but also in efficiency in creating the good produced. E.g., the entire success in terms of education development would be based on the efficient paradigm of the meaning of transforming the material in terms of money investment into tutoring profits.

Combined, the synthesis of results suggests that education spending in the form of public education in Delhi is a major factor in increasing education output, access, and quality. There are, however, inefficiencies, demographic pressures and lagged effects, which hint at a complex and multidimensional relationship. Though we need more expenditure to improve educational performance, it cannot be done by itself. The success of the expenditure is closely dependent on the allocation, management, and matching of the resources to the structural broader conditions. Thus, the conclusions draw attention to the necessity of a more sophisticated policy that would focus not only on the greater investment but also on the less wasteful and more efficient allocation of the resources and more prudent targeting.

5.2 Efficiency Interpretation

The efficiency analysis is a critical assessment of whether the augmented public investment in education in Delhi can be translated into a similar augmentation in the educational outcomes. Although the regression findings are systematically described as statistically significant and an affirmative connection between education expenditure and other key indicators of the process, e.g. the educational outcomes, enrolment, and student-to-teacher ratio (PTR), the efficiency aspect brings about a more subtle interpretation.

The outcome model results show that the coefficients of education expenditure are a strong positive coefficient ($\beta_1 \approx 4.32$). Similarly, in the enrolment model find that the elasticity is positive ($\beta_1 \approx 0.824$) and, consequently, a rise in expenditure will cause an increase in schooling. The PTR model also contributes to this trend, as the ratio of expenditure is negative

($\alpha_1 < 0$), which implies that as funding increases, this PTR is reduced, hence enhancing the quality at the classroom level. All these results are indicative that government expenditure is beneficial in improving various aspects of the education system.

Effectiveness, however, does not necessarily mean efficiency. The efficiency model, especially using the Stochastic Frontier Analysis due to the Stochastic Frontier Analysis framework (SFA) separates the impact of a random shock (v_t) and the impact of inefficiency (u_t). The fact that the level of inefficiency term is not zero means that they cannot solely attribute all the observed progress in results to optimum resource use⁴⁵. Virtually speaking, although Delhi shows an increase in outcomes as spending increases, the distance to the efficiency frontier implies that the same level of outcomes may be obtained at a lower level of expenditure or there may be a better level at which the same level of expenditure brings about an increased level of outcomes.

Additional indication of poor efficiency can be seen in the role of the control variables in the outcome model. Whereas school capital formation has a positive, yet comparably low coefficient (≈ 0.28), government final consumption expenditure has a negative one (≈ -0.80), indicating potential misallocation, or inefficiency in overall government spending. Also, the population ratio has a very high negative coefficient (≈ -33.45), which suggests that population pressures are hugely driving out the sincerity of expenditure. This suggests that a growth in expenditure without efforts to counter the structural constraints, like population growth, could result in declining returns.

The growth model adds another layer to the efficiency discussion. Despite the positive impact that education spending has on Gross State Domestic Product (GSDP) this effect is not instant and would be more pronounced when lagged variables are taken into consideration. The above lag structure indicates that returns to education spending occur over time, which supports the notion that short-run measures of efficiency can understate long-run payoffs. However, the revelation that capital formation has an exceeding impact on growth as opposed to education expenditure shows that the relative productivity of education spending can be less than that of other sectors.

In a sense to the point that even the low-order automatic correlation (also present in both the enrolment model and the PTR model, but smoothed out by the AR(1) adjustment), suggests that this limits the models as far as the accuracy of the business is considered because the model

is determined by the small scale. This statistical weakness indicates that the relationships estimated, though significant, may not be completely inclusive of sources of inefficiency or externalities.

In general, the discussion implies that the field of social spending on education in Delhi is both efficient and ineffective. Though the increase in expenditure is bound to cause measurable alterations in the outcomes, accessibility and quality, the impact of inefficiency, limitations of the demographics, and quite low impact on economic development give evidence that resources are not used most efficiently. The current policy focus, therefore, needs to be amended not only to increase spending, but also the allocation and utilisation as well. These include better focusing of funds, institution structures and the necessity to overcome structural constraints that limit the productivity of educational investments.

5.3 Growth Linkages

Education expenditure and economic growth in Delhi is an example of a multi-stage transmission mechanism, as although public spending affects economic growth, it does so directly and indirectly because of the improvement in educational outcomes. Empirical findings gained by the growth model show that education does not have a direct, unconditional, and direct effect on economic performance; instead, it has indirect effects that are lagged, conditional and channelled through human capital formation.

The approximation equation shows that the direct impact of education expenditure (EDU_EXP) is positive but rather moderate, indicating that growth is not necessarily proportional when the expenditure on education is increased. Conversely, education results (EDU_OUTCOME) show a more significant and statistically significant relationship with GSDP, especially when they are lagged ($EDU_OUTCOME_{t-1}$ to $EDU_OUTCOME_{t-4}$). This time delay effect validates the argument that economic returns to education do not come instantaneously because gains on learning outcomes manifest in a slow process of improvement in the labour market.

In addition, the analysis shows that capital formation is the strongest variable of economic growth, as its coefficients are substantially greater than educational ones. This result implies that education as a determinant of growth, although it does have an influence, it does so in a complementary manner, as it accumulates physical capital, not as a substitute. These findings

emphasize a structural attribute of the Delhi economy, in which growth remains mostly investment-based, with human capital in a supportive manner.

The resultant analysis shows that the indirect route of influence, which is represented by the mediation framework, is stronger than the direct impact. In particular, school-level outcomes are enhanced with education spending and, consequently, economic growth. This should be understood to mean that effectiveness in scaling the expenditure into quantifiable results is a critical factor towards growth effectiveness⁴. That is, more spending with economic payoffs is only meaningful in cases where the spending results in real changes in the quality and performance of education.

Nevertheless, the sample size is quite small, which limits the time series, providing some limitations. The fact that mild autocorrelation persists and the degrees of freedom are small can influence the accuracy of estimates of the coefficients. Also, since other more structural variables (technological change or labour market dynamics) are absent, the model is only able to represent the growth process at its partiality. Hence, although the findings provide strong evidence that there is an education-growth relationship, they are not to be interpreted without further consideration.

On the whole, the results demonstrate that education does not act as a short-run stimulus, but a long-term one. The success of state expenditure in stimulating economic growth does not only rely on the amount of money that has been spent, but on how well it enhances the educational performance and its compatibility with the larger economic frameworks. This highlights why the shift in policy priorities towards the growth in spending must change to efficiency results from the outcomes of investments in the education sector, so that investments in education yield long-run economic returns.

5.4 Comparison with Literature

The empirical evidence of the present paper generally concurs with the prevailing body of literature that has also found the positive impact of public spending on education outcomes, but has also identified key peculiarities of the efficiency and transmission strategies. The findings reveal that the expenditure on education positively and significantly ($p = 0.0357$) influences school-level achievements, but the estimated effect ($\beta_1 \approx 4.32$) is noteworthy. This agrees with a substantial amount of empirical research, which determines that public

investment improves learning outcomes, especially when it is improved by infrastructure and institutional support.

The results are, however, not limited to merely an outcome improvement but a multidimensional aspect in terms of access, quality, and efficiency. The enrolment model shows good and significant elasticity ($\beta_1 \approx 0.824$), showing that more expenditure will be an effective factor in fostering school attendance. This is in line with current knowledge that financial allocation will reduce entry barriers. However, the existence of weak autocorrelation, even post-AR(1) adjustment, implies that enrolment dynamics are somewhat path-dependent, which common past studies based on cross-sectional data have underestimated.

Qualitatively, the PTR model indicates that expenditure has a negative and significant correlation with pupil/teacher ratio, meaning the more money is spent, the better the classroom situations are. This is in line with the theoretical expectations but contrary to certain empirical studies, which pose weak or insignificant relationships between spending and quality indicators³⁷. This may be explained by the fact that the impact of initially relatively target-oriented spending on the teacher recruitment and infrastructure in the city of Delhi is greater.

The efficiency analysis is more critical from a point of view. Although outcomes, enrolment, and quality factors get better with spending, the model of DEA/SFA shows that efficiency is not always optimal, meaning that increases in spending do not have a recommended proportionate output. This aligns with the efficiency literature, which frequently highlights diminishing returns and institutional inefficiencies in public sector spending.

Lastly, the growth model brings in a macroeconomic dimension in that education is an influencing factor to GSDP, but with a lagged impact, whereas capital formation is the predominant driver of growth. This relationship is further narrowed down by the mediation framework that shows education outcomes as a mediating channel between expenditure and growth. This layered finding adds to the literature by empirically validating the indirect pathway, which is often theoretically assumed but less frequently quantified.

On the whole, the paper validates the beneficial role of education spending, yet incisively brings out the fact that effectiveness and efficiency are not interchangeable and thus adds a more integrated and refined view to the existing literature.

5.5 Policy Implications

The study's empirical data shows that the study has a statistically significant positive impact on education in Delhi, using the school-level achievement, enrolment and quality of education, using the Pupil-Teacher Ratio (PTR). However, the efficiency analysis shows that more spending is not always optimal in any way, which makes them use resources in a more resolved and conscious way.

The spending on education in Delhi has gone up significantly over the years, with the education sector contributing to around 20-25 per cent of the state budget in recent years, one of the highest among the Indian states. Despite this, the existence of inefficiencies as illustrated by the DEA/SFA framework demonstrates that improvement in outcome in response to an increase in spending is disproportional. This emphasizes how the expenditure should be specifically directed, especially to where the marginal returns are at their highest point. As an example, the regression results have shown that school capital formation has a positive but a relatively lower coefficient than direct education expenditure, implying that although infrastructure investment is exciting, its effect is maximized when it is combined with the improvement of the quality of instruction and the use of resources.

Moreover, the negative and significant correlation between education expenditure and PTR suggests that greater expenditure has led to better teacher supply. Practically, Delhi has cut PTR in government schools to amounts of about 30:1 at the primary level, which is the Right to Education (RTE) average, but the differences occur at the interdistrict level³⁵. This demands a quality-oriented policy, and expenditure is not channelled to employment of teachers only, but also towards improvement of teacher training, tracking of learning performance and eliminating intra-regional inequalities.

Importantly, the findings of the growth model indicate that the effects of education on economic growth are lagging and are mediated by the way they promote better outcomes. Thus, the policy needs to be oriented more towards long-term efficiency-focused planning instead of a short-term expansion of expenditures, where financial inputs would be methodically connected to quantifiable educational outputs and larger developmental targets.

CHAPTER 6: CONCLUSION

6.1 Summary

The current paper focuses on exploring the dependence and level of the current level of educational results of in-school learning in Delhi, and how the pattern and level of the existing government expenditure on education correlates with the estimation of a multi-model econometric model that has been estimated using EViews. It is computed based on time series information (approximately 15 values), which is combined with an outcome, access, quality, efficiency and growth dimension to provide the multidimensional view of the efficiency in expenditure on education.

The findings indicate that the impact on the outcome of the government spending on education on the performance of the school-based outcomes is significant and positive. In the first regression model, the higher the cost of education, the higher the outcome of education, which can be measured in terms of positivity, as the coefficient estimates are significant (e.g., 0.43). However, this relationship does have its moderating factors (population pressure and government consumption), which are negatively associated, so that an increase in spending would not have correlations associated with certain proportional increases.

Access-wise, the enrolment model indicates that education spending has a positive impact on school involvement, and the values of the elasticity of education spending are close to 0.8. The use of AR(1) correction does decrease the autocorrelation, but some weak persistence is still present, which is the result of a small sample size and potential inertia in the enrolment trends⁴². This indicates that though financial inputs effectively increase access, the increase is slow rather than instant.

This Pupil-Teacher Ratio (PTR) will be reflected as the aspect of quality that will be given in our case, since the more the expenditure, the higher the quality of education, as this ratio will decrease. The hypothesis is proven by the negative and statistically significant coefficient, according to which an even distribution of funds in the form of personnel and facilities may give better classroom conditions, and, therefore, learning conditions.

The efficiency analysis is performed using the tools of frontier-based methods (DEA/SFA) and indicates that the accuracy of converting financial resources into education products is not even at its best, even with higher spending. An efficiency score of less than unity will reveal the

existence of both allocative and technical inefficiencies, which means that given the same resources, it could have better performance due to an increase in resource utilization skills and focus.

At the macroeconomic level, the growth model forms a lagged relationship between education and GSDP, indicating that growth in educational attainment would not result in economic growth immediately but occur over time. Capital formation turns out to be a more potent immediate source of growth, and education has an indirect and lagging impact. This can be further supported by the mediation model, which affirms that the education results are a transmission channel whereby the public expenditure leads to the economic performance.

In general, the paper finds that though government spending in Delhi has a positive influence on education access, quality, and results, its efficacy is only partial and its effect on economic growth is indirect and time-dependent, but not direct.

6.2 Key Conclusions

Empirical evidence which can be summarized as follows: At least the statistically significant effect of the amount of funds spent on education in the City of Delhi is observed as to the level of school-level outcomes; the quality of expenditure is not the same measure of every aspect of education performance. The main regression estimates show that education spending has a big positive impact on educational outcomes, where its coefficient value in the primary model is in the range of above 4.0, indicating that further increases in expenditure have a significant impact of about 4.0 or more on outcomes that can be measured⁴³. Likewise, the enrolment model validates that an increase in public expenditure strongly boosts school attendance, and the elasticity is estimated at 0.82, but the autocorrelation was still weak, which is attributed to structural constraints, probably because of the lack of years of observation of the time-series.

The pupil-teacher ratio (PTR) model is another model that shows an increase in the quality as the number of instances of spending in the model increases. The fact that the coefficient is negative and significant implies that the larger the amount of money is, the lower the PTR and this negatively affects the situation within a classroom. But it is the rate of growth which shows the diminishing marginal returns to this growth, i.e. there is a threshold beyond which additional expenditures will lead to inconsequential increases in quality.

Although there are some positive associations, the examination of the efficiency reveals a more subtle conclusion. DEA/SFA structure suggests that the level of expenditure may be relatively high, and the respective scores of efficiency do not equal at the optimum level, which means that there may be a level of resources that are translated to results inefficiently. This is not equivalent to the input intensity to the output realisation, whereby there are disparities in assigning and using processes.

Lastly, the growth model lays out that the effects of education on economic performance do not occur instantaneously, and there is a time delay. The addition of lagged education indicates that an increase in human capital is a contributor to increasing Gross State Domestic Product (GSDP) over the long period of time, and capital formation becomes a stronger contributor in the short term. The results, as a whole, confirm the fact that even though the increase in educational outcomes is a necessary condition associated with the increase in public spending, this condition is not sufficient to guarantee the efficiency, or the maximisation of the long-term economic returns.

6.3 Recommendations

The results of the study posit that although the statistical significance of the effects of the increase in public spending on education in Delhi on the school-level results and school enrolment and quality measures, including the pupil teacher ratio (PTR), have improved, the effectiveness of this spending has not been balanced. Policy recommendations, therefore, have to be directed not only at hiking expenditure but also at enhancing how the funds are allocated and utilised.

The need to redefine spending to result dimensions comes next. The empirical data also reveal that education expenditure (EDU_EXP) represents such a strong positive inducer of change that it influences the education outcomes, but the situation is very pathetic that the impact of other factors, such as the government final consumption, is very negligible and raises the question of ineffectiveness of the latter. As it is in the current fashion of budgets, Delhi is budgeting to spend about 20-25 percent State budget on education that emerges to be one of the best in India. At the marginal returns, there will be sub-optimal components of some of these parts. A higher proportion of the money that should be used in capital formation in schools, i.e. modernization of the infrastructure, digital learning, as well as other facilities, is

then proposed to be utilized which was found to have a more correlated relation to the results in a positive manner.

Second, teacher resource strengthening is one of the quality-enhancing mechanisms that should be prioritised. The findings of the PTR model suggest that there is a significant decrease in PTR with an increase in expenditure, which enhances the conditions in the classroom. This change, however, depends on effective teacher deployment and retention. To an average of 25-30 students per teacher, the level of PTR in the Delhi government schools has risen, but there are still disparities among districts⁴⁰. Policy should instead be concerned with specific teacher recruitment, subject matter specialization and lifelong professional education, instead of the blind increase. Also, the long-term benefits of reallocation of funds into teacher development programs can outweigh the administrative expenditure that is generalized.

Lastly, there is inefficiency as supported by the estimates of DEA/SFA, and this means that there should be improved governance and monitoring mechanisms. Transparency in the use of funds, a relationship between spending and quantifiable performance indicators, can greatly enhance the efficiency of outcomes.

6.4 Future Research

The current study might be enhanced in future research in terms of the limitations and the scope of analysis. To start with, analysis is carried out on a rather small time-series data set with about 15 annual observations, which limits the statistical strength and restricts the possibilities to identify long-term structural relationships. The data should be further developed into a panel format, i.e., by adding additional Indian states over the next 20-25 years, so that the number of observations (which can be up to 400-600 data points) increases considerably and can be more confidently estimated with the help of a fixed (or random effects) model.

Second, a cross-state comparative analysis would make it possible to better evaluate efficiency differentials in education spending. Delhi is a high-expenditure area, and therefore may have different outcome dynamics than less-expensive states like Bihar or Uttar Pradesh. A comparison of efficiency scores, outcome elasticities and other quality metrics across states would be a more effective way of addressing whether increased spending systematically relates to improved educational performances.

Moreover, the next generation of research might also incorporate dynamic panel models to capture more appropriately the lagged impacts of education results on economic growth⁴¹. This would capture the temporal aspect in a more effective manner compared to the current methodology, which is limited in its observations and may have autocorrelation problems. On the whole, more general data sets and comparative frameworks would improve the validity and policy relevance of the discoveries.

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